

What is an Operating System?

An operating system (OS) is system software that manages computer hardware and provides common services to application programs. It acts as an interface between users, applications, and hardware such as the CPU, memory, storage, and input/output devices.

The main goals of an operating system are efficient resource management, convenient program execution, security, and coordination between software and hardware.

Major Responsibilities

Process Management: The OS creates, schedules, synchronizes, and terminates processes. A process is a program in execution, and the scheduler decides which ready process receives CPU time.

Memory Management: The OS tracks memory usage and allocates memory to processes. Virtual memory allows secondary storage to extend the apparent amount of available main memory.

File Management: The OS organizes data into files and directories and manages creating, reading, writing, deleting, and protecting files.

Device Management: The OS coordinates hardware devices through device drivers and provides applications with standardized interfaces for devices such as keyboards, displays, printers, and storage controllers.

Kernel

The kernel is the central component of an operating system. It manages privileged operations including CPU scheduling, memory management, device access, and system calls. Applications normally request OS services through system calls rather than directly controlling hardware.

Types of Operating Systems

Batch Operating Systems: Process groups of jobs with limited interactive user involvement.

Time-Sharing Operating Systems: Allow multiple users or tasks to interact with a computer by rapidly switching CPU time among processes.

Real-Time Operating Systems: Designed for applications where responses must be produced within specified timing constraints. They are common in robotics, industrial control, automotive systems, and embedded devices.

Distributed Operating Systems: Coordinate resources across multiple networked computers and attempt to provide a unified computing environment.

Embedded Operating Systems: Designed for specialized devices with limited resources or specific operational requirements.

Common Operating Systems

Linux: An open-source operating-system family widely used on servers, cloud infrastructure, development systems, supercomputers, and embedded devices.

Windows: A widely used operating-system family for personal computers and enterprise environments, providing a graphical desktop, hardware support, security features, and application compatibility.

macOS: A Unix-based desktop operating system designed for Mac computers and closely integrated with Apple's hardware and software ecosystem.

Android: A Linux-based mobile operating-system platform designed primarily for smartphones, tablets, televisions, and other connected devices.

Process Scheduling

Common CPU scheduling approaches include First-Come, First-Served (FCFS), Shortest Job First (SJF), Priority Scheduling, and Round Robin. Round Robin assigns each ready process a time quantum and is commonly associated with interactive systems.

Memory Management

RAM provides fast storage for currently active programs and data. The operating system tracks which memory regions are available and which are allocated to processes.

Virtual memory separates addresses used by programs from physical memory locations. Paging divides virtual memory into fixed-size pages and physical memory into corresponding frames. A page that is not currently in RAM can be stored on secondary storage and loaded when needed.

File Systems

A file system defines how files and directories are organized and stored on a storage device. File systems commonly maintain metadata such as file names, sizes, timestamps, permissions, and storage locations.

Examples include ext4, NTFS, APFS, and FAT32. Different file systems provide different features for permissions, journaling, compatibility, storage capacity, and reliability.

Operating System Security

Operating systems use authentication and authorization mechanisms to control access to resources. Authentication verifies a user's identity, while authorization determines what an authenticated user or process is allowed to access.

Other security mechanisms include process isolation, memory protection, access permissions, secure boot, encryption support, and security updates.

System Calls

A system call is a controlled interface through which a user-level program requests a service from the operating system kernel. Examples include creating processes, opening files, reading data, writing data, and inter-process communication.

Suggested Test Questions

English: What are the main responsibilities of an operating system?

Hindi: Operating system mein memory management kya karta hai?

Hinglish: Round Robin scheduling kaise kaam karta hai?

Cross-lingual: Linux kis type ka operating system hai?

Specific: Which page discusses file systems?

No-context: What is the market price of Linux in 2026?